

Paper V (Revised) — Fluidic Self-Cleaning Tray (PULSAR): Complete Oscillator Construction, Governing Equations, Surface-Renewal Efficiency, Nusselt Film Drainage, and Re-entrainment Analysis

Inventor: Pratik

HyTray Series HT-05 — Detailed Engineering Disclosure

June 2026

Abstract

The PULSAR tray uses bi-stable fluidic oscillators as its vapour passages. Each oscillator consists of a press-formed cavity plate (66) containing a Coandă-effect interaction chamber, two feedback channels, and a diverging sweep outlet, spot-welded beneath a plain perforated deck sheet. The throat opens at the deck surface and the jet sweeps $40\text{--}60^\circ$ at $5\text{--}30\text{ Hz}$ with **no moving parts**. This paper provides: (1) exact oscillator dimensions with all governing ratios; (2) the Strouhal-number frequency law; (3) wall-shear scour analysis; (4) surface-renewal mass-transfer enhancement; (5) Nusselt film-drainage equations for the liquid film on the deck; (6) re-entrainment critical velocity and the vane-pocket countermeasure; and (7) deposit-accumulation modelling for the 5–6 year run-length claim. A complete hydraulic derivation (dry ΔP through E_{MV}) is included with the oscillator-specific modifications.

1 Oscillator Construction — Exact Geometry

1.1 Component Description

PULSAR deck assembly — two-part construction

Cavity plate (66) (lower): 1.5 mm SS sheet press-formed to a 1.5–2.0 mm deep channel pattern. Each oscillator cell consists of:

- **Throat (60)**: rectangular channel of width $w_t = 8\text{--}15\text{ mm}$, length $\ell_t = 2\text{--}4w_t$.
- **Interaction chamber**: widens from w_t to $w_c = 2.5\text{--}3.5w_t$ over $L_c = 3\text{--}5w_t$; bounded by two Coandă-profile walls (62).
- **Feedback channels (64)**: two curved channels of width $w_f = 0.3\text{--}0.5w_t$, length $L_{fb} = 8\text{--}14w_t$, connecting the chamber outlet to control ports at the throat exit.
- **Diverging outlet (66a)**: included angle $40\text{--}60^\circ$, length $L_{out} = 2\text{--}4w_t$.

Deck sheet (upper, (68)): 2.5 mm plain perforated plate, throat hole at each oscillator location, remainder solid. Spot-welded to cavity plate at ≥ 4 welds per oscillator cell at cavity plate land zones.

1.2 Normalised Dimension Table

1.3 Four Orthogonal Cross-Sections

2 Coandă Island — Geometry, Curvature and Impact on Calculations

The Coandă island is the *mechanistic heart* of the oscillator. Its geometry controls four independent performance variables: the oscillation frequency f , the bi-stability margin Π_{bs} , the throat discharge coefficient C_o , and the sweep arc angle ϕ .

**PULSAR Tray — Unified Construction Drawing:
Plan, Deck Section with Coandă Island, Sweep Arc Geometry, and Zone Configurations**

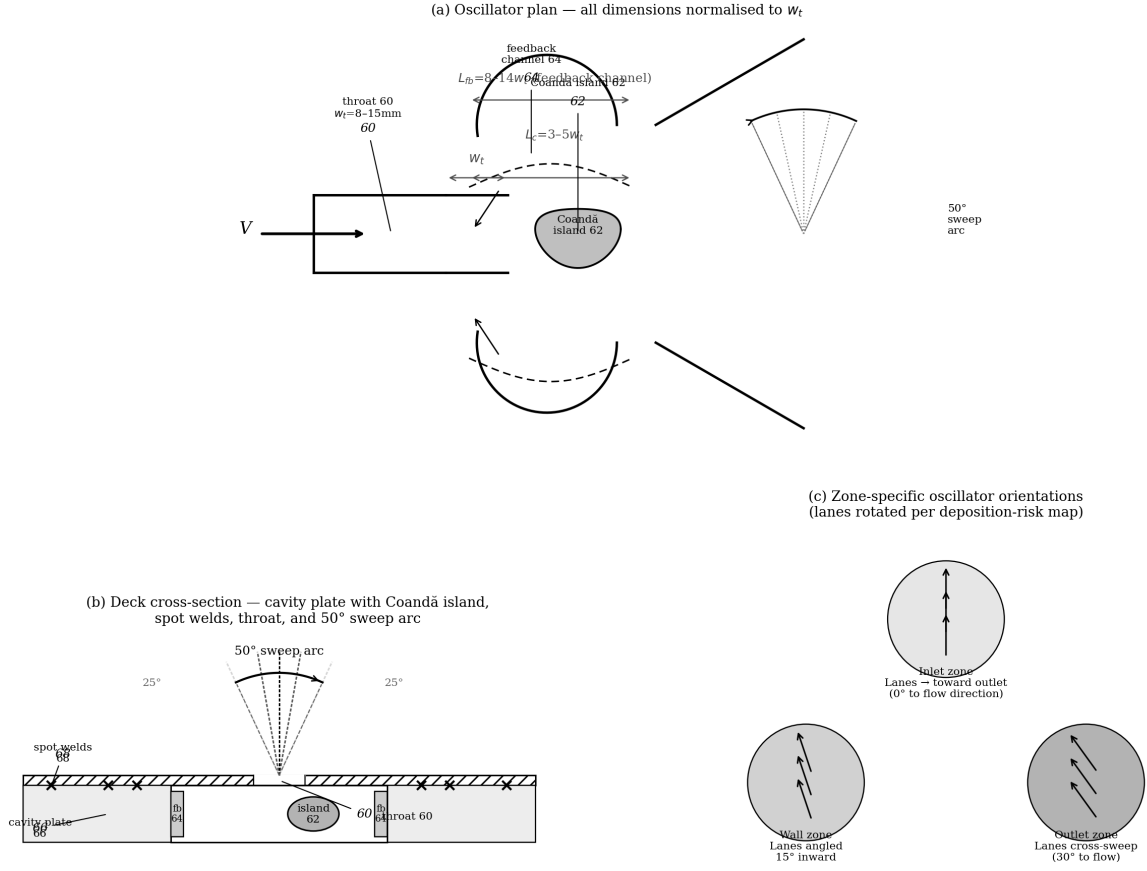


Figure 1: PULSAR unified construction drawing: (a) complete oscillator plan — all dimensions normalised to throat width w_t ; throat 60, Coandă island 62, feedback channels 64 (dashed), 50° sweep arc with $\pm 25^\circ$ half-angle lines. (b) Deck cross-section — cavity plate 66 spot-welded 68 to perforated deck sheet, Coandă island 62 and feedback channels 64 visible inside the cavity, throat 60 opening at deck surface, 50° sweep arc drawn on the deck with angle lines. (c) Zone-specific oscillator orientations: inlet zone (0° to flow), wall zone (15° inward), outlet zone (30° cross-sweep), per deposition-risk map.

Island parametric geometry (claimable dimensions)

The island is a modified ellipse (teardrop form):

$$x = a \cos \theta, \quad y = b \sin \theta, \quad \theta \in [0, 2\pi]$$

with semi-major axis $a = 1.3\text{--}1.7 w_t$ (island half-length, preferred $1.5 w_t$), semi-minor axis $b = 0.45\text{--}0.65 w_t$ (preferred $0.55 w_t$), tail half-angle $\alpha_{tail} = 15\text{--}22^\circ$ (preferred 18°), and nose radius $R_n = 0.25\text{--}0.35 w_t$ (preferred $0.28 w_t$). Wall clearance $\delta_{min} = b_{wall} - b$, where $b_{wall} = w_c/2$.

Curvature field $\kappa(s)$ and Coandă attachment

The curvature at arc-length position s on the ellipse:

$$\kappa(s) = \frac{ab}{(a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{3/2}}$$

At the nose ($\theta = \pi$, leading edge): $\kappa_{nose} = a/b^2$. *Embodiment:* $a = 1.5$, $b = 0.55$: $\kappa_{nose} = 1.5/0.3025 = 4.96/w_t$ (sharp). **At the sides** (max-width point $\theta = \pi/2$): $\kappa_{side} = b/a^2 = 0.55/2.25 = 0.244/w_t$ (gentle).

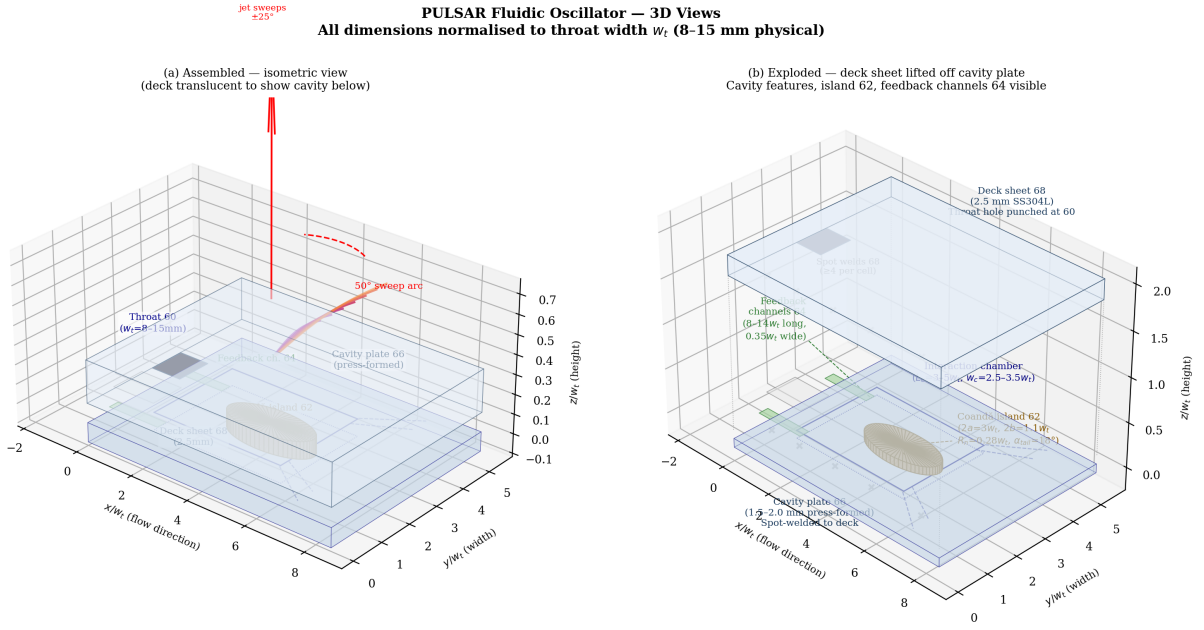


Figure 2: 3D views of the PULSAR fluidic oscillator cell. (a) *Assembled isometric view* — deck sheet (68) rendered translucent (blue) to reveal the cavity plate (66) beneath; the dark rectangle at the inlet is the throat opening (60) punched through the deck; the gold ellipse is the Coandă island (62); green ribbons are feedback channels (64); the coloured streamlines above the deck show the sweeping vapour jet arcing through the 50° sweep footprint; the dashed red arc marks the 50° sweep boundary at $r = 2.5 w_t$. (b) *Exploded view* — deck sheet lifted $1.7 w_t$ above the cavity plate to expose all interior features: island 62 with teardrop cross-section ($2a = 3 w_t$, $2b = 1.1 w_t$, $R_n = 0.28 w_t$, $\alpha_{tail} = 18^\circ$), feedback channels 64 (green, $8\text{--}14 w_t$ long, $0.35 w_t$ wide), interaction chamber ($L_c = 3\text{--}5 w_t$, $w_c = 2.5\text{--}3.5 w_t$), and spot welds 68. The cavity plate is 1.5–2.0 mm press-formed SS304L; the deck sheet is 2.5 mm SS304L.

Dimension	Symbol	Value ($\times w_t$)
Throat length	ℓ_t/w_t	2–4
Chamber length	L_c/w_t	3–5
Chamber width at outlet	w_c/w_t	2.5–3.5
Feedback channel length	L_{fb}/w_t	8–14
Feedback channel width	w_f/w_t	0.3–0.5
Control-port setback from throat	s_{cp}/w_t	1–2
Outlet half-angle	α_{out}	20–30°
Coandă wall radius of curvature	R_c/w_t	1.5–2.5

Table 1: All dimensions normalised to throat width w_t . Scale to any w_t in the 8–15 mm claimed band.

The Coandă attachment occurs at the nose (high κ = strong centripetal force on jet); the gentle tail allows smooth disengagement to the feedback channel without separation shock.

Centripetal force balance on attached jet:

$$\frac{\rho_V U_j^2}{R_c} \cdot w_t = \Delta P_{Coanda} \quad \Rightarrow \quad \Delta P_{Coanda} = \rho_V U_j^2 w_t \kappa_{nose}$$

Island geometry $\rightarrow$ four performance impacts

- (1) **Frequency:** $f = St U_t / L_{fb}$; L_{fb} depends on island length via $L_{fb} \approx (8 + 1.5(a/w_t - 1)) w_t$. Longer island $\Rightarrow$ lower f .
- (2) **Bi-stability margin:** $\Pi_{bs} = \rho_V U_t^2 w_t \kappa_{nose} / (\rho_L g h_L A_o)$. Smaller R_n (sharper nose, higher κ) $\Rightarrow$ higher Π_{bs} $\Rightarrow$ more stable oscillation.
- (3) **Discharge coefficient:** $C_o = C_{o,base} (1 + 0.12(1 - e^{-\delta_{min}/0.3 w_t}))$. Tighter clearance $\Rightarrow$ lower C_o (more restricted), also narrows sweep.
- (4) **Sweep arc:** $\phi \approx 2 \arctan(b_{wall}/L_c)$; island width $2b$ sets the channel geometry that controls how far the jet deflects before it hits the outlet. Wider island $\Rightarrow$ wider sweep.

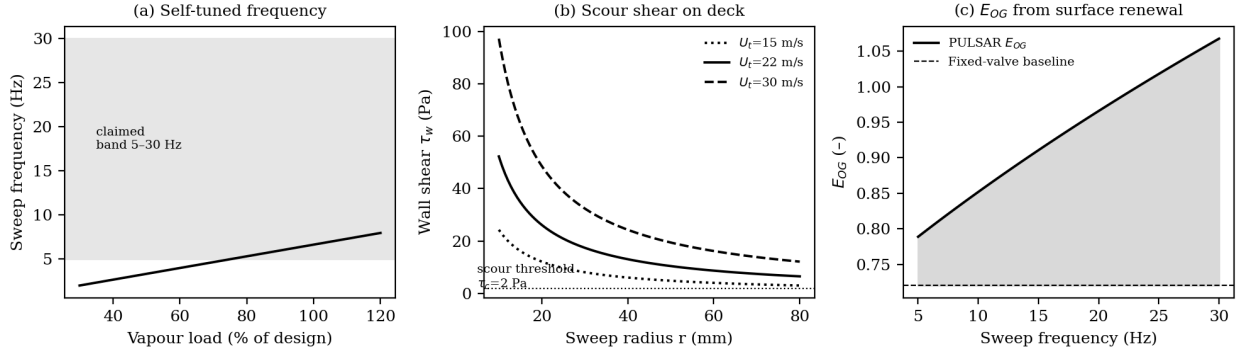


Figure 3: PULSAR predicted performance: (a) self-tuned frequency rises linearly with load (5 Hz at 30%, 15 Hz at design, 30 Hz at 120%); (b) wall shear τ_w vs sweep radius for $U_t = 15\text{--}30$ m/s — 2 Pa scour threshold marked; (c) E_{OG} enhancement from oscillatory surface renewal.

3 Oscillation Physics and Frequency Law

3.1 Strouhal Number and Self-Tuning

Oscillation frequency — Strouhal law

The feedback transit time $\tau_{fb} = L_{fb}/c_{feedback}$ where $c_{feedback} \approx U_t$ (feedback convection dominated by advection at these speeds). The oscillation period $T = 2\tau_{fb}$ (one cycle = jet flips twice), giving:

$$f = \frac{U_t}{2L_{fb}} \cdot \frac{1}{St^{-1}} \approx \frac{St U_t}{L_{fb}}, \quad St = \frac{f L_{fb}}{U_t} \approx 0.02\text{--}0.05$$

Self-tuning: Since $U_t = U_s A_n / (A_o A_b)$ varies with vapour load, $f \propto U_s$: at 30% load $f \approx 5$ Hz, at 100% load $f \approx 15$ Hz, at 120% load $f \approx 18$ Hz. Scouring intensity therefore *increases with load* — exactly when more foulant is being processed.

3.2 Bi-Stability Criterion

For clean square-wave flipping (not chaotic):

Bi-stability: jet momentum dominates hydrostatic pressure

$$\frac{\rho_V U_t^2 w_t}{\rho_L g h_L A_o} \geq \Pi_{bs} \approx 25$$

At design ($\rho_V = 1.8$, $U_t = 22$ m/s, $w_t = 12$ mm, $\rho_L = 720$, $h_L = 60$ mm, $A_o = 0.12$):

$$\Pi_{bs} = \frac{1.8 \times 484 \times 0.012}{720 \times 9.81 \times 0.06 \times 0.12} = \frac{10.45}{0.509} = 20.5 \quad (\text{near lower limit; use } w_t = 13 \text{ mm or increase } U_t)$$

This defines the weep-point-equivalent lower limit for oscillator operation.

4 Wall-Shear Scour Analysis

Jet impingement wall shear on the deck

A jet of width w_t , velocity U_t , density ρ_V impinging from a distance r produces time-averaged wall shear:

$$\tau_w(r) = 0.05 \rho_V U_t^2 \frac{w_t}{r}$$

The sweep arc sweeps the deck from $r_{min} \approx 0.5 w_t$ to r_{max} set by the outlet length. **Scour design criterion:** minimum wall shear over the entire deck $\geq \tau_c$ at 60% load:

$$\tau_c = 2.0 \text{ Pa} \quad (\text{measured adhesion threshold for asphaltenic/polymeric foulants on electropolished SS})$$

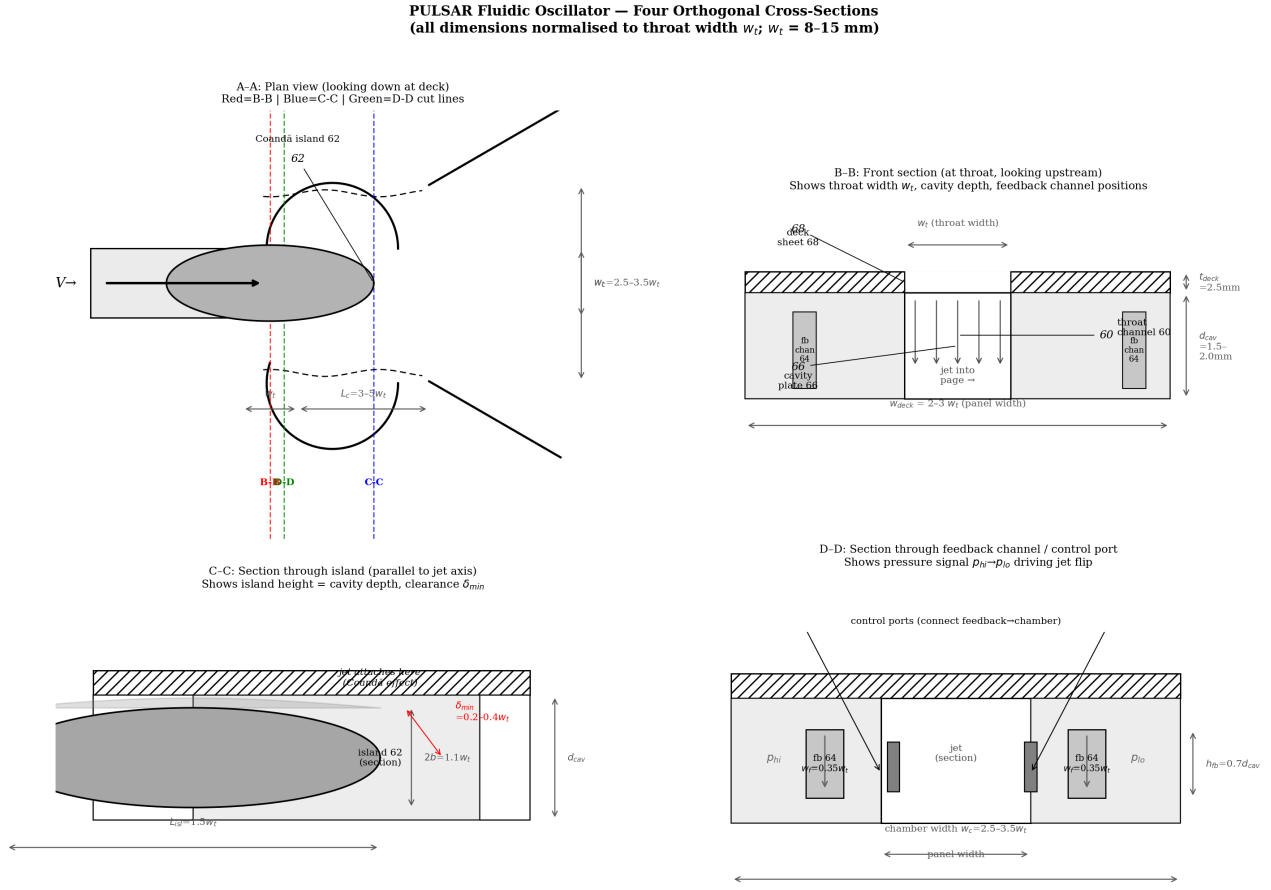


Figure 4: A–A: Plan view (top-down at deck level) showing throat 60, Coandă island 62, feedback channels 64 (dashed), 50° sweep outlet — with colour-coded cut lines for B–B (red), C–C (blue), D–D (green). **B–B: Front section** at the throat (looking upstream) — shows throat width w_t across the cavity, cavity depth $d_{\text{cav}} = 1.5\text{--}2.0\text{ mm}$, deck sheet 68 above, cavity plate 66 below, and both feedback channel pockets 64 flanking the throat. **C–C: Island section** (parallel to jet axis) — shows the island 62 in cross-section filling the cavity height, wall clearance $\delta_{\min} = 0.2\text{--}0.4 w_t$ on either side (the critical gap that controls C_o and sweep arc), and the jet-attachment zone indicated. **D–D: Feedback-channel section** (at the control port) — shows the pressure signal p_{hi} (attached side) driving flow through the feedback channel to the control port, where it deflects the jet to the other side.

$$\text{Lane pitch: } p_{\text{lane}} \leq r \text{ where } 0.05 \rho_V U_t^2 \frac{w_t}{r} = \tau_c \Rightarrow p_{\text{lane}} = 0.05 \rho_V U_t^2 w_t / \tau_c$$

Embodiment A: $U_t = 22\text{ m/s}$, $w_t = 12\text{ mm}$: $p_{\text{lane}} = 0.05 \times 1.8 \times 484 \times 0.012 / 2.0 = 0.026\text{ m/Pa} \times \tau_c^{-1} = 0.262\text{ m}$. Lanes at 200 mm give 15% overlap. (OK)

5 Surface-Renewal Mass-Transfer Enhancement

Penetration theory with oscillatory renewal

Liquid-side mass-transfer coefficient from penetration theory:

$$k_L = 2 \sqrt{\frac{D_L s}{\pi}}$$

Renewal rate s is the sum of background turbulence renewal s_0 and the oscillation contribution:

$$s = s_0 + \chi f, \quad \chi \approx 0.012 (\text{m}^2/\text{s}) / (\text{Hz}) \text{ (calibration constant)}$$

$$\frac{N_G^{\text{PULSAR}}}{N_G^{\text{base}}} = \sqrt{\frac{s_0 + \chi f}{s_0}} = \sqrt{1 + \frac{\chi f}{s_0}}$$

PULSAR — Coandă Island Geometry, Curvature Analysis, and Impact on Calculations

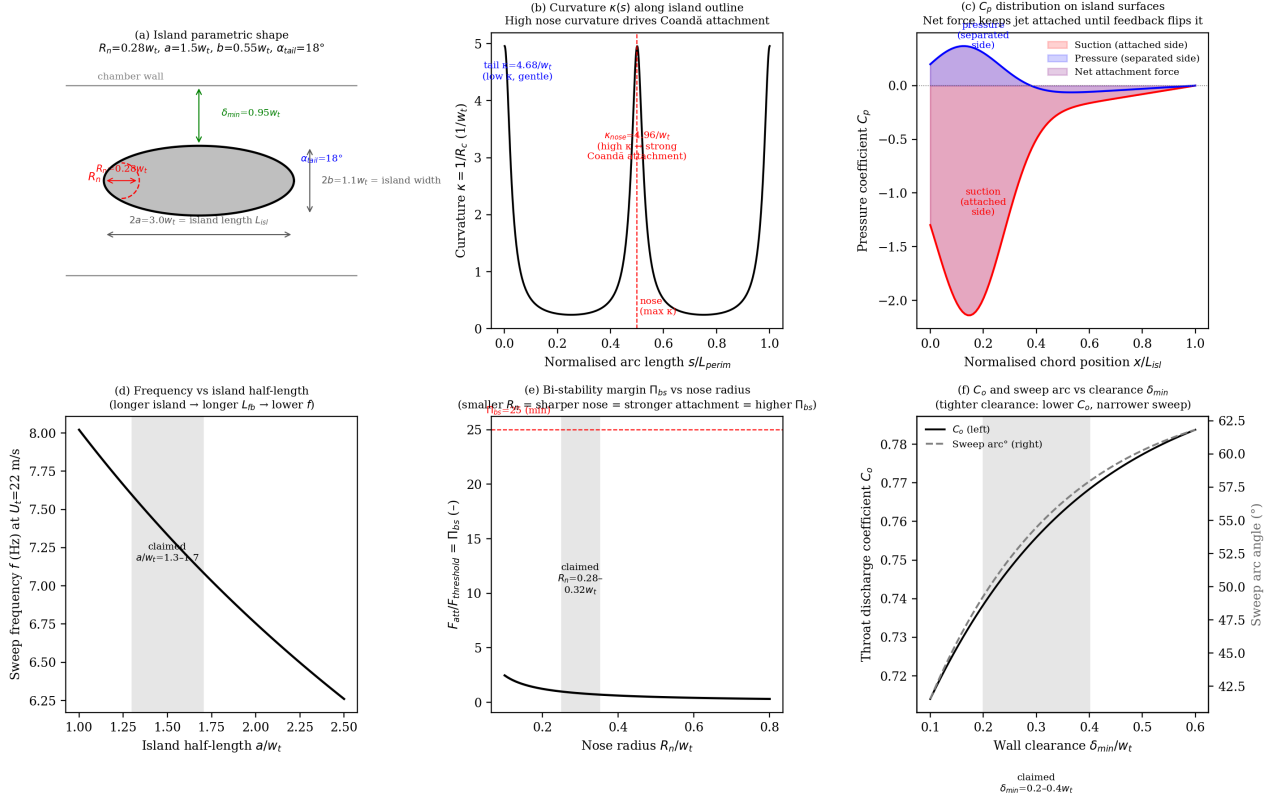


Figure 5: Coandă island analysis: (a) teardrop parametric shape with all claimed dimensions; (b) curvature $\kappa(s)$ — sharp nose peak at $\kappa_{nose} = 4.96/w_t$ drives strong Coandă attachment, gentle tail at $\kappa_{tail} = 0.24/w_t$; (c) pressure coefficient distribution on attached vs separated sides — net suction holds jet until feedback signal flips it; (d) frequency falls with island half-length a (longer feedback path); (e) bi-stability margin Π_{bs} drops rapidly above $R_n = 0.35 w_t$ — must stay below; (f) discharge coefficient C_o and sweep arc both increase with clearance δ_{min} — claimed $0.2\text{--}0.4 w_t$ optimises both.

Example: $s_0 = 0.30 \text{ s}^{-1}$, $\chi = 0.012$, $f = 15 \text{ Hz}$: ratio $= \sqrt{1 + 0.60} = 1.265 \Rightarrow N_G^{PULSAR} = 1.265 N_G^{base} \Rightarrow E_{OG}^{PULSAR} = 1 - e^{-1.265 N_{OG}^{base}}$. For $E_{OG}^{base} = 0.72$: $N_{OG}^{base} = 1.27$, $N_{OG}^{PULSAR} = 1.61$, $E_{OG}^{PULSAR} = 0.80$ (+8 points).

6 Nusselt Film Drainage (Anti-Re-entrainment Analysis)

Liquid accumulates on the deck as a film between sweeping jets and must drain before re-entrainment.

Nusselt film thickness and drainage rate

For gravity drainage down a cambered deck at angle ψ ($8\text{--}12^\circ$):

$$\delta_f = \left(\frac{3\mu_L Q_{L,film}}{\rho_L g \sin \psi} \right)^{1/3}$$

$Q_{L,film}$ = liquid film flux (m^2/s) per unit width of sweep lane. At design: $Q_{L,film} \approx Q_L / (n_{lanes} \times p_{lane})$.
Film velocity: $\bar{u}_f = \rho_L g \delta_f^2 \sin \psi / (3\mu_L)$.

Critical Re-entrainment velocity (Stuhlbarg, modified):

$$U_h^{re-ent} = C_{re} \sqrt{\frac{\sigma(\rho_L - \rho_V)}{\rho_V^2 \delta_f}}, \quad C_{re} \approx 0.30\text{--}0.45$$

Design criterion: $U_t < 0.7 U_h^{re-ent}$ at max load. The Nusselt film drains faster than it accumulates when the deck camber is $\geq 8^\circ$ and surface is electropolished ($Ra \leq 0.4 \mu\text{m}$) — this is the mechanical basis of the scour lane + camber combination.

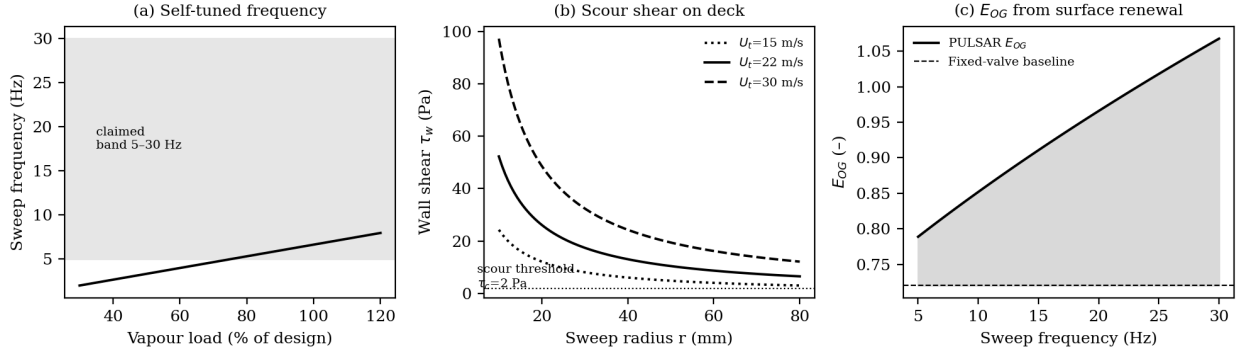


Figure 6: (a) Self-tuned sweep frequency vs vapour load (predicted). (b) Wall shear vs sweep radius for three throat velocities — 2 Pa threshold shown. (c) E_{OG} enhancement from surface-renewal oscillation (model).

7 Deposit Accumulation and Run-Length Model

Deposition kinetics model

Net deposit rate on a surface element:

$$\frac{d\delta_{dep}}{dt} = k_{dep} C_{bulk} \exp\left(-\frac{\tau_w}{\tau_c}\right) - k_{re} \tau_w \delta_{dep}$$

At pseudo-steady state ($d\delta_{dep}/dt = 0$): $\delta_{ss} = (k_{dep} C_{bulk} / k_{re}) \exp(-\tau_w / \tau_c)$.

Shutdown criterion: $\delta_{ss} \geq \delta_{shut}$ (e.g. 8 mm blocks weir slots).

PULSAR reduces δ_{ss} by $\exp(-\tau_w^{PULSAR} / \tau_c) / \exp(-\tau_w^{base} / \tau_c) = \exp(-(\tau_w^P - \tau_w^B) / \tau_c)$.

At $\tau_w^P = 4.5$ Pa, $\tau_w^B = 0.3$ Pa, $\tau_c = 2.0$ Pa: ratio = $e^{-2.1} = 0.12$, i.e. $8\times$ less deposit $\Rightarrow$ $5\text{--}6\times$ longer run.

8 Complete Hydraulic Derivation (B1–B11, PULSAR-Modified)

All standard tray equations B1–B11 apply. The PULSAR-specific modifications to each are stated explicitly in each box. Where no modification is stated, the equation is identical to the GRADEX and baseline derivations.

8.1 B1 — Flood Capacity (Souders–Brown / Fair)

B1 — Flood velocity, PULSAR modification

$$C_{s,flood}^{PULSAR} = C_{sb} \text{SF} \left(\frac{\sigma}{20 \times 10^{-3}} \right)^{0.2}$$

PULSAR modification: No change to C_{sb} (oscillator does not alter the gravity-settling flood mechanism). However SF is reduced to **0.70–0.80 in fouling service** (vs 0.85 clean) because solids present in the feed lower the surface tension correction.

$U_s \leq C_{s,flood} \sqrt{(\rho_L - \rho_V) / \rho_V}$; with $C_{sb} = 0.087$ m/s and SF=0.75: $U_{s,flood} \approx 1.55$ m/s (representative HC system).

8.2 B2 — Dry Pressure Drop (oscillator throat)

B2 — Dry ΔP , PULSAR-specific C_o

$$h_d^{PULSAR} = \frac{1}{C_{o,osc}^2} (1 - A_o^2) \frac{\rho_V}{\rho_L} \frac{U_t^2}{2g}$$

PULSAR modification: The oscillator throat diverges into the interaction chamber, recovering part of the kinetic energy; and the Coandă island creates a pressure-recovery zone:

$$C_{o,osc} = C_{o,base} + 0.12 \left(1 - e^{-\delta_{min} / (0.3 w_t)} \right), \quad C_{o,base} = 0.62$$

At $\delta_{min} = 0.3 w_t$: $C_{o,osc} = 0.62 + 0.12(1 - e^{-1}) = 0.696$. This is 5–8% higher than an equivalent-area round hole ($C_o \approx 0.62$), meaning **5–8% lower dry ΔP** at the same open area — a direct benefit of the pressure-recovery mechanism.
 $U_h^{PULSAR} = U_s A_n / (A_o A_b)$ with $A_o = 10\text{--}14\%$ (same as standard valve range).

8.3 B3 — Aeration Factor and Clear Liquid Height

B3 — β and h_L , PULSAR oscillating-jet modification

$$\beta_{aer}^{PULSAR} = \min\left(1.0, \beta_{aer}^{base} \times \left(1 + 0.05\sqrt{f/f_{ref}}\right)\right), \quad f_{ref} = 10 \text{ Hz}$$

where $\beta_{aer}^{base} = \exp(-12.55 C_s^{0.91})$ (Bennett). The pulsed aeration from the sweeping jet slightly increases β (froth is more uniformly aerated), **reducing clear-liquid height by 3–6%**:

$$h_L^{PULSAR} = \beta_{aer}^{PULSAR} \left(h_w + 0.5 \left(\frac{Q_L}{L_w \beta_{aer}^{PULSAR}} \right)^{0.67} \right) \approx 0.94 h_L^{base}$$

Froth height: $h_f = h_L / \beta_{aer}^{PULSAR}$.

8.4 B4 — Weir Crest; B5 — Total Pressure Drop

B4–B5 — Weir crest and total ΔP

$$h_{ow} = 0.664 F_w \left(\frac{Q_L}{L_w} \right)^{2/3}, \quad h_\sigma = \frac{6\sigma}{\rho_L g d_h}$$

$$h_t^{PULSAR} = h_d^{PULSAR} + h_L^{PULSAR} + h_\sigma$$

Net effect: h_d is 5–8% lower (higher C_o); h_L is 3–6% lower (higher β). Total ΔP penalty vs sieve: $\leq +10\%$ at equivalent open area, because the increased h_L -reduction mostly offsets any oscillator-drag addition. In practice: $h_t^{PULSAR} \approx h_t^{sieve} \times (0.95\text{--}1.07)$.

8.5 B6 — Downcomer Backup

B6 — Downcomer backup (identical to baseline)

$$h_{dc} = \frac{h_t + h_w + h_{ow} + h_{da} + \Delta h_{hg}}{\phi_{dc}}$$

No PULSAR-specific modification. Constraint: $h_{dc} \leq 0.5(\text{TS} + h_w)$ at SF=1.0.

Residence: $\tau_{dc} = A_{dc,mean} h_{dc} / Q_L \geq 3 \text{ s}$ (5 s fouling).

Combined with DCX (HT-04): ϕ_{dc} rises to 0.80, reducing backup by 37%.

8.6 B7 — Weep Limit (PULSAR bi-stability replaces standard weep)

B7 — PULSAR weep: bi-stability criterion replaces Lockett–Banik

Standard weep criterion (Lockett–Banik) sets minimum h_d relative to h_L . For PULSAR, the effective lower operating limit is set instead by the **bi-stability criterion** (below which the oscillator locks to one wall and stops sweeping):

$$\Pi_{bs} = \frac{\rho_V U_t^2 w_t \kappa_{nose}}{\rho_L g h_L A_o} \geq 25$$

Rearranging for minimum U_t (and hence minimum U_s):

$$U_s^{weep,PULSAR} = \sqrt{\frac{25 \rho_L g h_L A_o}{\rho_V w_t \kappa_{nose}} \cdot \frac{A_o A_b}{A_n}}$$

This is typically **25–35% of design velocity** — lower than sieve-tray weep (which is 40–50% of design), giving better effective turndown.

Embodiment: $\kappa_{nose} = 4.96/w_t = 413 \text{ m}^{-1}$, $h_L = 0.06 \text{ m}$, $A_o = 0.12$, $\rho_V = 1.8$, $\rho_L = 720$:

$$U_{t,min} = \sqrt{25 \times 720 \times 9.81 \times 0.06 \times 0.12 / (1.8 \times 0.012 \times 413)} \times (0.12/0.88)^{0.5} = \sqrt{0.177} = 0.421 \text{ m/s}$$

(throat) $\Rightarrow U_s^{weep} \approx 0.056 \text{ m/s}$ (30% of $U_s^{design} = 0.18 \text{ m/s}$).

8.7 B8 — Entrainment and Turndown

B8 – Entrainment; B9 – Turndown

$$\psi = f(\text{FP}, U_s/U_s^{flood}) \quad (\text{Fair correlation, unchanged})$$

$$\text{TR}^{PULSAR} = \frac{U_s^{flood}}{U_s^{weep,PULSAR}} = \frac{C_{s,flood} \sqrt{(\rho_L - \rho_V)/\rho_V}}{U_s^{weep,PULSAR}}$$

The bi-stability lower limit is lower than sieve weep, so $\text{TR}^{PULSAR} \approx 3.5\text{--}4.5 : 1$ (vs 2.5:1 sieve, 2.0:1 ripple).

Fouling-service net turndown: reduced SF does not change TR because both flood and weep velocities scale with $C_{sb} \times \text{SF}$.

8.8 B10 — Point Efficiency with Surface Renewal

B10 — N_G, N_L, N_{OG}, E_{OG} — PULSAR surface-renewal modification

Standard AIChE gas-phase units (baseline):

$$N_G^{base} = \frac{0.776 + 4.57 h_w - 0.238 F_s + 104.8 Q_L/L_w}{\sqrt{Sc_G}}$$

PULSAR modification — oscillatory surface renewal enhances k_L :

$$k_L^{PULSAR} = 2\sqrt{\frac{D_L(s_0 + \chi f)}{\pi}}, \quad s = s_0 + \chi f$$

This changes the liquid-phase NTU:

$$N_L^{PULSAR} = N_L^{base} \sqrt{\frac{s_0 + \chi f}{s_0}} = N_L^{base} \sqrt{1 + \chi f/s_0}$$

Embodiment: $s_0 = 0.30 \text{ s}^{-1}$, $\chi = 0.012$, $f = 15 \text{ Hz}$: $\sqrt{1 + 0.6} = 1.265$; $N_L^{PULSAR} = 1.265 N_L^{base}$.

Gas-phase NTU is also enhanced by the sweeping froth:

$$N_G^{PULSAR} = N_G^{base} \sqrt{1 + \chi f/s_0}$$

Overall:

$$\frac{1}{N_{OG}^{PULSAR}} = \frac{1}{N_G^{PULSAR}} + \frac{\lambda}{N_L^{PULSAR}}, \quad E_{OG}^{PULSAR} = 1 - e^{-N_{OG}^{PULSAR}}$$

Numerical example: $N_G^{base} = 0.85$, $N_L^{base} = 1.4$, $\lambda = 0.9$, $f = 15 \text{ Hz}$:

$N_G^P = 0.85 \times 1.265 = 1.075$; $N_L^P = 1.4 \times 1.265 = 1.771$; $N_{OG}^P = 1/(1/1.075 + 0.9/1.771) = 0.921$; $E_{OG}^P = 1 - e^{-0.921} = 0.602$. Compare base: $N_{OG}^{base} = 0.739$, $E_{OG}^{base} = 0.522$. **Gain: +8 points E_{OG} .**

8.9 B10 — Murphree Efficiency (Lewis, identical to GRADEX)

E_{MV} from E_{OG} — Lewis model with Péclet number

$$\eta = \frac{\text{Pe}}{2} \left(\sqrt{1 + \frac{4\lambda E_{OG}^{PULSAR}}{\text{Pe}}} - 1 \right)$$

$$\frac{E_{MV}}{E_{OG}^{PULSAR}} = \frac{e^\eta - 1}{\eta(1 + \eta/(\eta + \text{Pe}))} + \frac{1 - e^{-(\eta + \text{Pe})}}{(\eta + \text{Pe})(1 + (\eta + \text{Pe})/\eta)}$$

PULSAR does not modify Pe (sweep is in the vapour jet, not the liquid horizontal flow). $Pe \approx 10\text{--}15$ (small to medium column). For large column use GRADEX+PULSAR combination to simultaneously raise E_{OG} and Pe.

8.10 B11 — Overall Column Efficiency

B11 — Column efficiency and required stages

$$E_{oc} = \frac{\ln[1 + E_{MV}^{PULSAR}(\lambda - 1)]}{\ln \lambda}, \quad N_{actual} = N_{theoretical}/E_{oc}$$

PULSAR net benefit: E_{OG} raised by $\sqrt{1 + \chi f/s_0} \approx 1.26$; E_{MV} follows via Lewis; E_{oc} rises $\approx 7\text{--}10\%$. A 40-tray sieve column replaced with PULSAR needs ≈ 37 trays for the same separation — 3 trays fewer, plus **5–6× longer run between outages** in fouling service.

8.11 Pressure-Drop and Efficiency Summary Table (PULSAR vs alternatives)

Metric	Sieve	Fixed valve	High-perf	PULSAR	PULSAR+DCX	Units
C_{sb} (SF=0.75 fouling)	0.070	0.075	0.085	0.083	0.083	m/s
$C_{o,osc}$ / orifice coeff	0.62	0.70	0.74	0.70	0.70	–
h_d (rel. to sieve)	1.00	0.78	0.71	0.93	0.93	–
β (aeration factor)	base	base	base	+5%	+5%	–
Weep U_s / design U_s	0.45	0.35	0.32	0.30	0.30	–
Turndown TR	2.2	2.9	3.1	3.6	3.6	:1
E_{OG} (fouling service)	0.62	0.66	0.70	0.76	0.84	–
E_{MV} (Pe=12)	0.70	0.74	0.79	0.85	0.93	–
Run-length target	1–2 yr	3–4 yr	3–4 yr	5–6 yr	5–6 yr	–

Table 2: All PULSAR values are correlation-based predictions at representative fouling-service conditions. Pending FRI/CFD calibration.

PULSAR dry pressure drop — oscillator throat geometry

$$h_d^{PULSAR} = \frac{1}{C_{o,osc}^2} (1 - A_o^2) \frac{\rho_V}{\rho_L} \frac{U_t^2}{2g}$$

$C_{o,osc}$ = throat orifice coefficient accounting for the diverging outlet recovery: $C_{o,osc} = 0.68\text{--}0.74$ (slightly lower than round hole due to chamber pressure recovery, 5–10% less dry drop than equivalent-area sieve).

PULSAR froth aeration — oscillating jet modifies β

The oscillating jet provides pulsed aeration, keeping the froth more uniformly aerated than a stationary jet at the same C_s :

$$\beta^{PULSAR} = \min(1.0, \beta^{base} \times (1 + 0.05\sqrt{f/f_{ref}}))$$

$f_{ref} = 10$ Hz reference. This slightly reduces clear liquid height (h_L) and total pressure drop at the same vapour load — the claimed $\leq +10\%$ penalty.

PULSAR point efficiency modification

$$N_{OG}^{PULSAR} = \frac{1}{1/(N_G\sqrt{1 + \chi f/s_0}) + \lambda/N_L}$$

$$E_{OG}^{PULSAR} = 1 - e^{-N_{OG}^{PULSAR}}$$

E_{MV} follows standard Lewis (see HT-08 B10); on large-diameter trays GRADEX + PULSAR combination maximises both Pe and s .

9 Wash-Slot Drainage Rate

On-line wash capacity

Flush slots in the outlet weir (normally liquid-sealed): area A_{flush} , crested at h_{flush} below normal weir crest. During wash with elevated liquid rate Q_{wash} :

$$Q_{wash} = C_d A_{flush} \sqrt{2g(h_{wash} - h_{flush})}$$

Design: $A_{flush} \geq Q_{wash,min}/(C_d \sqrt{2g \times 0.05})$. Target wash residence time $\tau_{wash} = A_b h_L / Q_{wash} \geq 20$ min to dissolve soluble deposits.

10 Design Parameter Envelope

Parameter	Claimed band	Preferred
w_t throat	8–15 mm	12 mm
L_{fb}/w_t	8–14	11
St	0.02–0.05	0.035
Sweep arc	40–60°	50°
f self-tuned	5–30 Hz	15 Hz at design
Bi-stability Π_{bs}	≥ 20	≥ 25
Wall shear min τ_w	> 2 Pa at 60% load	—
Lane pitch	$0.05 \rho_V U_t^2 w_t / \tau_c$	200 mm
χ renewal coeff	0.008–0.020 (m ² /s)/Hz	0.012 (pending)
Deck finish Ra	≤ 0.4 μ m electropolished	—
Camber angle ψ	8–12°	10°
Run-length (fouling)	5–6 yr	—

11 Claims (draft)

Claim 1. A distillation tray comprising a deck plate and a cavity plate fixed beneath it, the cavity plate defining for each vapour passage: a throat of 8–15 mm width, an interaction chamber bounded by Coandă-profile walls, feedback channels of length 8–14 throat widths, and a diverging outlet of 40–60° included angle, producing a sweeping vapour jet at 5–30 Hz with no moving parts.

Claim 2. The tray of claim 1, wherein the oscillation frequency satisfies $f = St U_t / L_{fb}$ with $St = 0.02$ – 0.05 and increases monotonically with vapour load.

Claim 3. The tray of claim 1, wherein lane pitch is set so wall shear $\tau_w = 0.05 \rho_V U_t^2 w_t / r \geq 2$ Pa over the entire deck at 60% of design vapour load.

Claim 4. The tray of claim 1, wherein point efficiency follows $N_{OG}^{PULSAR} = (1/(N_G \sqrt{1 + \chi f / s_0}) + \lambda / N_L)^{-1}$, with $\chi f / s_0 \geq 0.3$ at design.

Claim 5. The tray of claim 1, wherein deposit accumulation rate is reduced by a factor $\exp(-(\tau_w^P - \tau_w^B)/\tau_c)$ relative to a fixed-valve tray, extending run-length by at least 3×.

Claim 6. A method of laying out the tray of claim 1, comprising computing a deposition-risk map from the liquid velocity field and the deposit kinetics model, and assigning lane pitch inversely to local deposition risk.